

Independent vs. Chain Pattern in Food-related Businesses, NYC

Exploring Urban Data with Machine Learning

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Research Question

- What neighborhood features are most predictive of whether chain or independent food businesses dominate a given area in NYC?
- To what extent do socioeconomic and built environment indicators explain the spatial distribution of chain vs. independent food businesses in NYC?

Dataset Used

POI (OpenStreetMap)

- Filtered to include food-related businesses, such as restaurants, cafes, fast food, bakeries
- Each POI tagged as either chain or independent by name frequency

NYC Census Tract Boundary

- Used as spatial units of analysis
- Aggregate with POIs and merging with socioeconomic data

ACS Census Variables

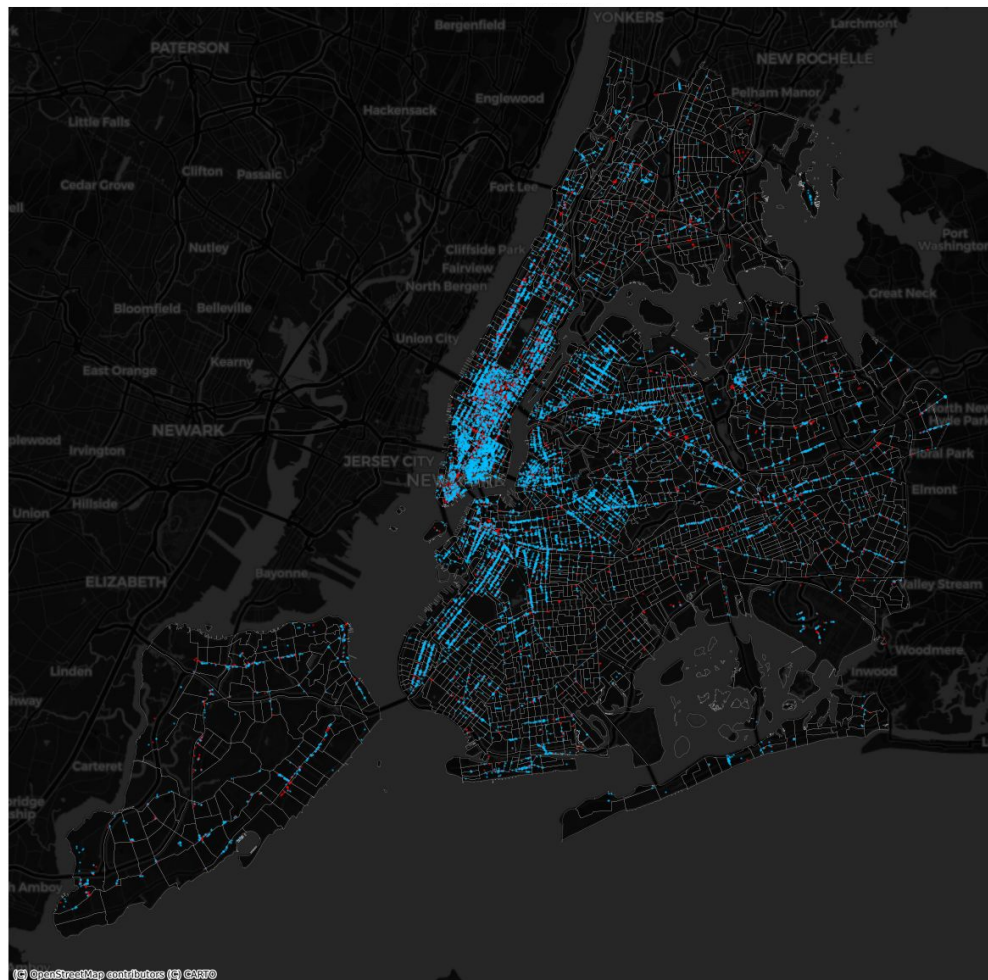
- Poverty rate (% of people below poverty line)
- Unemployment rate
- No-vehicle household rate

Zoning / Land Use Data

- % Residential land per tract
 - % Open Space land per tract
-

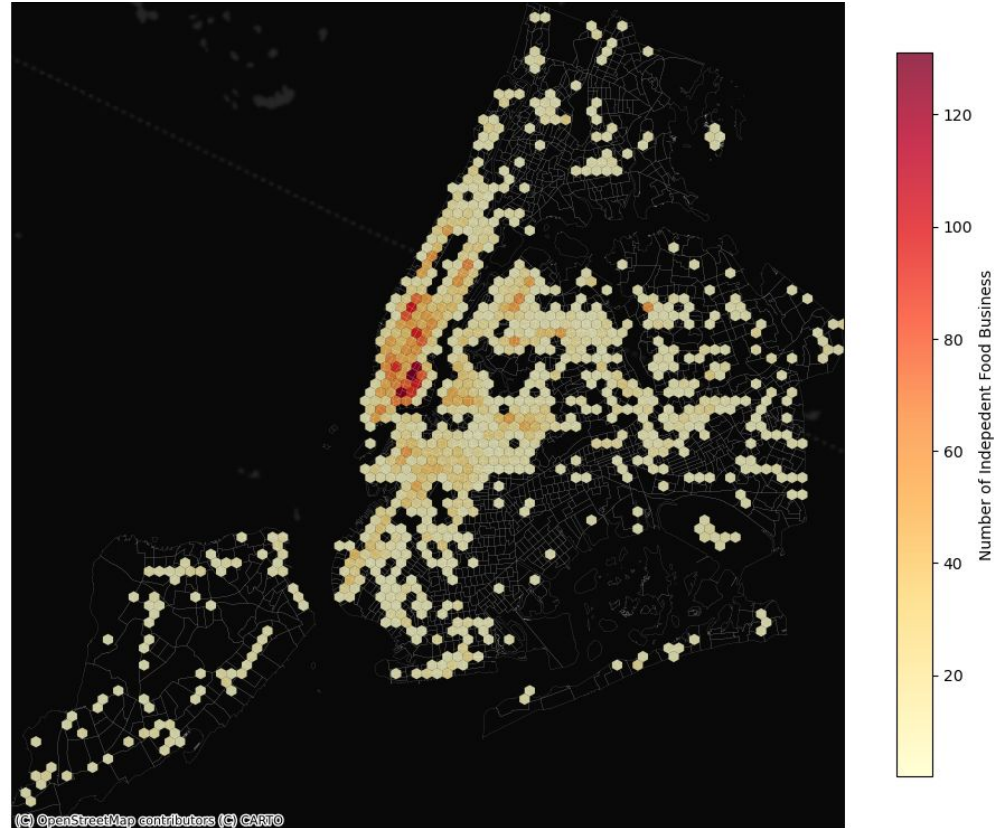
Food-Related POIs in NYC

- Blue: Independent & Red: Chain
- In food-related business, independent business dominate overall, mostly in
 - Lower Manhattan
 - West part of Brooklyn
 - Northwest Queens
- Chain business seems to be more concentrated in
 - Staten Island
 - South Bronx
- Urban core area has high density of POIS, but mostly indie-dominated



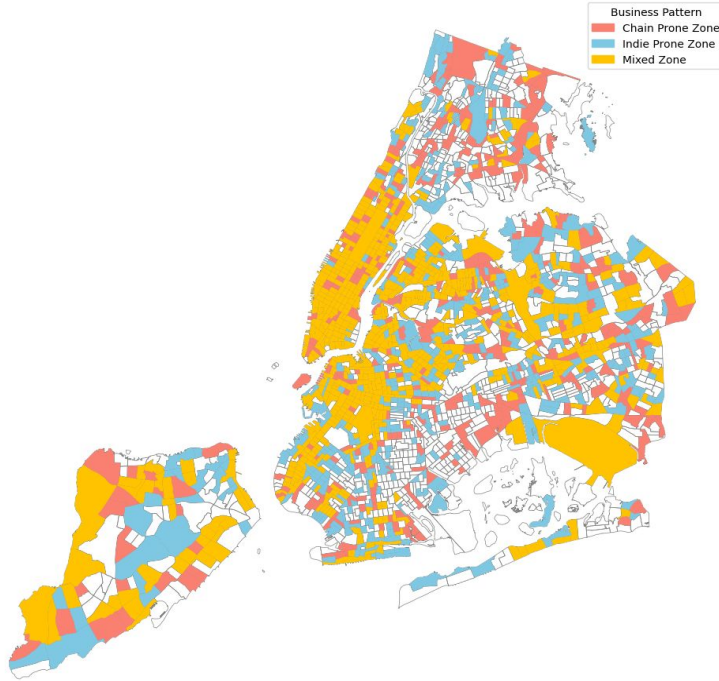
Independent Food Business Hotspots in NYC

- Strongest hotspots are concentrated in
 - Lower Manhattan
 - Williamsburg, Downtown Brooklyn
 - Parts of Astoria and Bushwick
- Weaker to no presence
 - Southern Brooklyn
 - Eastern Queens
 - Most of Staten Island



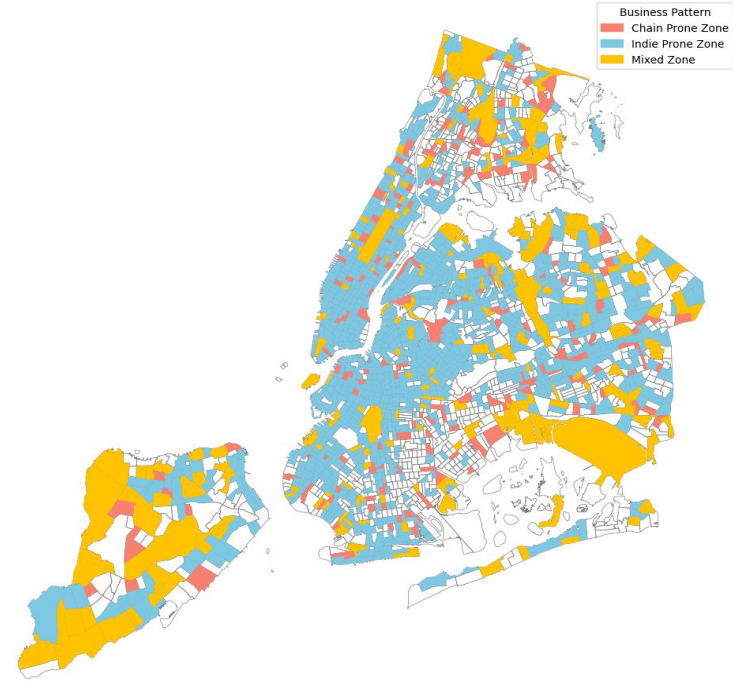
Classification for Food Store Typologies

Store Type Classification by Census Tract



- Calculated based on raw count: % of chain or independent dots in census tract

Store Type Classification (weighted) by Census Tract



- Classified based on store density (per km²)

Weighted Store Ownership Classification & Features

% Residential Land

- Dense residential tracts may favor small businesses?
- Indie clusters aligned with residential cores?

Poverty Rate

- High indie seems to presence in lower-income neighborhoods?

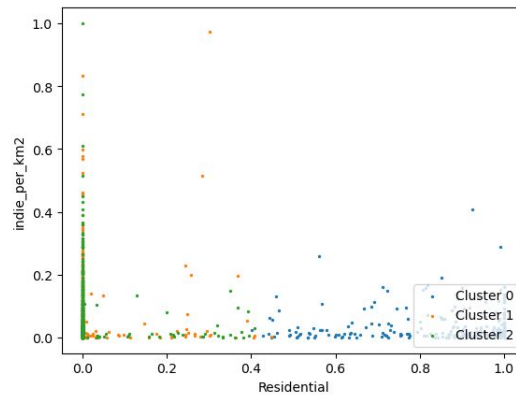
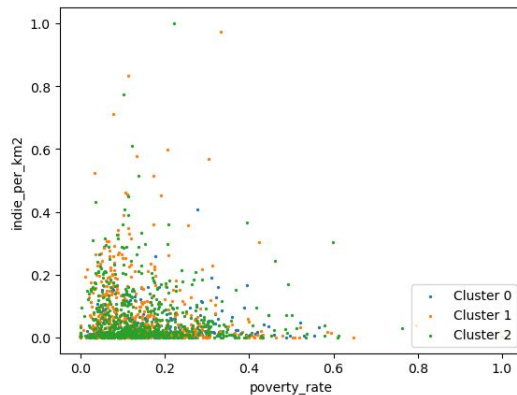
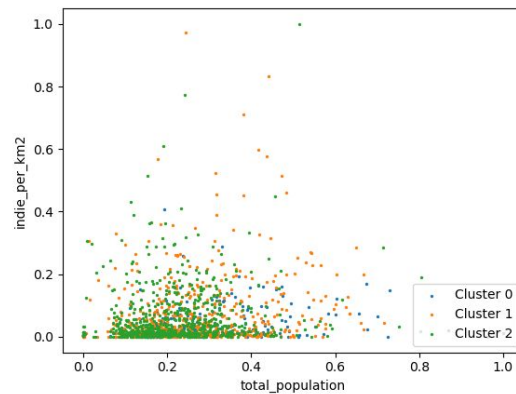
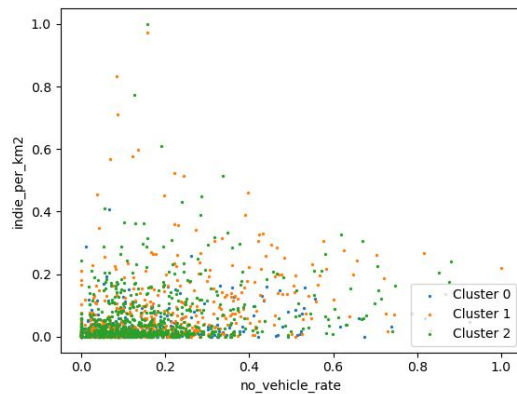
Population

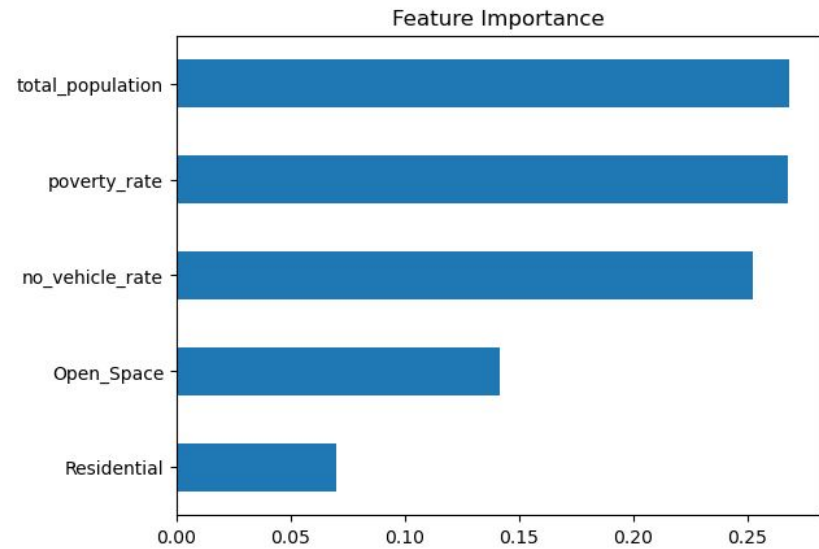
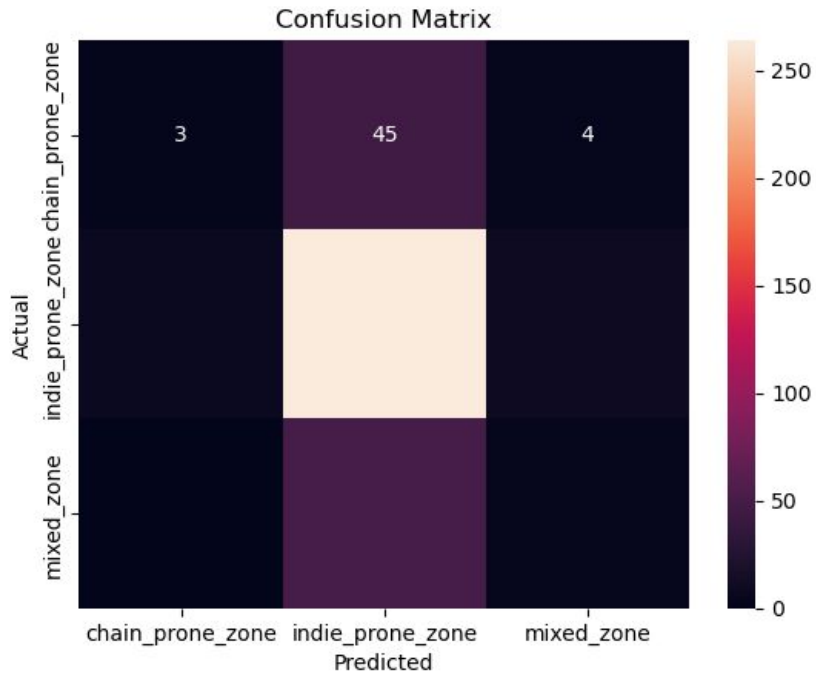
- Denser areas—especially indie-prone—had visibly more POIs?

% Households without Cars

- Indie hotspots clustered in car-light, transit-accessible neighborhoods?
-

K-Means Clustering: Exploring Feature Relationships with Indie Business Density





Random Forest Classifier

- Overall Accuracy: 70%
- F1-Score (Indie-prone): 0.83
- Indie-prone zones are easily distinguishable, while chain-prone and mixed zones are often misclassified.

- Accuracy: 0.70

Confusion Matrix (Actual\Predicted)

	Chain	Independent	Mixed
Chain	1	51	0
Independent	10	263	7
Mixed	3	44	6

K-Nearest Neighbor Classifier

Training set score: 0.726

Test set score: 0.727

	feature	odds_ratio
0	total_population	0.999921
1	poverty_rate	1.009632
2	no_vehicle_rate	1.000767
3	Open_Space	0.917465
4	Residential	1.029972

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[[ 0 52  0]
 [ 0 280  0]
 [ 0 53  0]]
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- The model shows high overall accuracy but only because it defaults to predicting the dominant class.
- Odds ratios reveal very mild influence of any individual variable—none stand out as strong linear predictors across all classes.

Logistic Regression Classifier

Limitation

- The zone labels used as classification targets were not from a verified retail typology database
- Class imbalance strongly impacted model performance—further tuning or resampling could help.
- Future studies could:
 - Incorporate temporal data (e.g., openings/closures over time)
 - Add more detailed land use zoning regulations as explanatory variables
 - Maybe zoom in to single borough with more sparse store distribution will improve the predictability
 - Compare NYC patterns with other U.S. cities

Implications

The findings may suggest that independent businesses are not randomly scattered—they concentrate in specific types of neighborhoods shaped by mobility, density, and affordability.

- Urban planners and policymakers aiming to support small businesses should focus on:
 - Enhancing walkability and transit access in lower-income, high-density areas
 - Recognizing and preserving mixed-use residential districts with organic independent food clusters
 - Avoiding one-size-fits-all incentives

Conclusion

This project began with a simple question: Where do independent food businesses thrive in NYC, and what factors shape their spatial patterns? By combining clustering, mapping, and supervised learning, I found that indie-prone zones form recognizable spatial patterns may tied to features like population density, poverty, and transit reliance, but not necessarily strongly relevant.

Bibliography

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THANK YOU